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Non-prescription pain relievers

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Pharmacist CPE

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Regulars

Pain can have a significant impact on a person's quality of life including their physical and mental health, and ability to work, sleep, exercise and perform everyday tasks.

See page 06, Facts Behind the Fact Card: Non-prescription pain relievers

Recommending the most appropriate pain reliever

By John Bell, Self Care Principal Adviser

In community pharmacy, there's not a day goes by when, as pharmacists and pharmacy assistants, we're not asked for a pain reliever. Pain is probably the most common symptom with which we are presented; and so pain relievers are the most commonly requested products.

Very frequently, however, our customers ask for a pain reliever by brand name; they're preconditioned to request a specific product. Usually this preconditioning has occurred as a result of the weight of advertising; sometimes it's on the recommendation of a friend or family member; and sometimes it's because of previous positive history with the product.

Even so, we should never assume the product requested is the one likely to be the most suitable for our customer's needs.

The type of pain, how it occurred, what other medicines have been taken for the pain, what other conditions the customer might have, and what (if any) medicines are being taken for these conditions will all influence the recommendation we make for a pain reliever. Sometimes no over-the-counter pain reliever might be suitable. Referral or non-medication therapy might be a better option.

Of course, the two most commonly used non-prescription pain relievers – paracetamol and ibuprofen – are available from locations other than pharmacies (supermarkets, grocery stores, newsagents, service stations and so on); but that does not reduce our responsibility to ensure our customer is getting the right product and knows exactly how to take it to get the maximum benefit. In fact, accepting this responsibility much more clearly demonstrates the difference between pharmacy and non pharmacy outlets.

This issue of *inPHARMation* provides us with the principles of pain management with non-prescription medicines.

Generally pain can be described as acute or chronic. Acute pain starts suddenly and usually (but not always) has a readily identifiable cause, is of short duration and is self limiting. Sports injuries, toothache and period pain are examples. This is the kind of pain which tends to diminish as the body heals and can be alleviated with OTC pain relievers.

Chronic pain (that is pain which is long lasting or persistent) may be related to medical conditions such as arthritis or cancer, or it can also develop from acute pain if the acute pain is not effectively managed in the first place. So-called neuropathic pain (pain associated with nervous system injury) is a common cause of chronic pain. Psychological and emotional factors often contribute to chronic pain and the symptoms of chronic pain should be assessed and managed by a doctor. In some cases referral to a hospital pain clinic might be necessary.

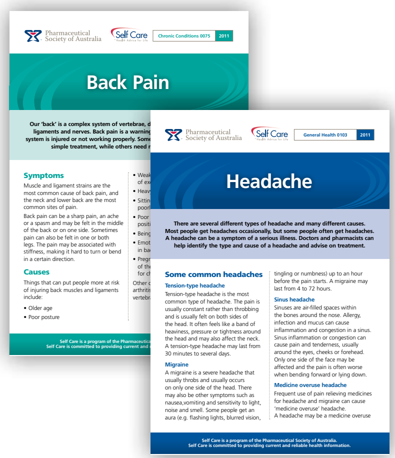
The non steroidal anti-inflammatory medicines (NSAIDs) available without prescription are listed with their recommended doses in Table 2 on page 18 of Counter Connection. They may be preferred to paracetamol where inflammation is associated with the pain. Studies have shown that in recommended non-prescription doses and for the recommended non-prescription duration (i.e. up to three days) ibuprofen has similar gastrointestinal tolerability to paracetamol (and need not be taken with food); however, as with all NSAIDs, there are some specific precautions which need to be considered. (See Practice Point 3.) Customers requesting an NSAID with any of these 'risk factors' or taking prescription medicines should be referred to the pharmacist.

Table 3 describes an effective protocol for a pharmacy assistant in the management of requests for a pain reliever.

Non-prescription pain relievers containing codeine (i.e. in combination with aspirin, ibuprofen or paracetamol) are now *Pharmacist-Only* medicines; so requests for these products will need to be referred to the pharmacist. Nevertheless, it is interesting to note that the strength of codeine in the S3 pain relievers is not always sufficient to be effective; and not everyone finds codeine effective at any strength.

John Bell, Self Care Principal Adviser.

There is no doubt, pain management can be troublesome for our customers, whether the pain is acute or chronic; so it's a challenge for us as well. But, having asked the right questions and with a little thought and consideration, we can help; and that long suffering customer will become a very loyal one.



Electronic delivery

The John Bell Health Column is available weekly by email.

If your pharmacy would like to receive the column, please send your email details to psc.nat@psa.org.au

The service of advice

Pharmacies must leverage their most valuable asset – the service of advice – to compete with the increasingly 'easy to find' health sections of supermarkets.

Consumer marketing expert and lecturer, Dr Gary Mortimer told delegates at Australian Pharmacy Professional (APP) Conference that providing information is the most important factor in building consumer trust and loyalty.

'Providing information is the biggest driver of trust,' he said.

Dr Mortimer cited a literature review which found that the higher the level of counselling the higher the level of trust.

'Quality advice builds trust which leads to long lasting profitable customer relationships. Selection and training is likely to be the most important investment for enduring pharmacy success.

'Ensuring that your staff is equipped to give the right advice is critical for you and your pharmacy,' he said.

Dr Mortimer said that the buying pattern with pain is that people go to the supermarket to top up – because next week they might have a headache – but as soon as it becomes more complex they head for a pharmacy.

With the need for quality advice being one of the reasons people visit the pharmacy instead of the supermarket Reckitt Benckiser which sponsored Dr Mortimer's presentation, launched its QUM education kit for pharmacy assistants. The kit is based on Quality Use of Medicines principles and follows the idea of the right drug, right dose, for the right duration, ensuring that consumers get the best possible outcome. It aims to educate pharmacy assistants on quality use of OTC analgesics such ibuprofen and paracetamol.

Dr Gary Mortimer

PSA/Guild working group formed

PSA and the Pharmacy Guild have established a Professional Programs Working Group.

The Working Group aims to deliver quality professional pharmacy services through the community pharmacy network by developing and implementing the range of professional practice programs under the Fifth Community Pharmacy Agreement (5CPA).

The Working Group consists of members from the Guild and the PSA, and its role is focussed on:

- Progressing the professional practice programs in the 5CPA
- Enhancing professional practice programs continued from the 4CPA
- Providing input into the design, implementation and roll-out of these programs.

The first meeting of the Working Group was held on 22 February. Two further meetings are planned for later in the year.

At its meeting on the 22 February, the Working Group discussed:

- Pharmacy participation in the Pharmacy Practice Incentives Program (PPI Program)
- The uptake of new programs such as the PPI Program and the pilot phase of the MedsCheck/Diabetes MedsCheck Program
- Consideration of an implementation plan for the national roll-out of MedsCheck and Diabetes MedsCheck from 1 July 2012
- The effects of recent and upcoming changes to the Home Medicines Review Program and how they impact on the pharmacy profession
- Rural Pharmacy Programs and proposed changes to eligibility criteria
- Uptake of Clinical Interventions and Staged Supply education modules.

Men's health website

Andrology Australia has revamped its website at: www.andrologyaustralia.org to give men easier access to information on male reproductive health and associated conditions. The new website reflects a slight change in Andrology Australia's branding. The redesign aimed to keep the functionality of the original website, but to make it easier for consumers and health professionals to access a range of education and health promotion resources on male reproductive health and associated conditions. It provides greater interactivity with the addition of videos, and allows users to easily share the content via social media and email.

Insulin use increases

More than 220,000 Australians began using insulin to treat diabetes between 2000 and 2009, according to an Australian Institute of Health and Welfare (AIHW) report. The web-based report, Incidence of insulin-treated diabetes in Australia, 2000–2009, shows that 77% of these people had Type 2 diabetes, 12% had gestational diabetes and 10% had Type 1 diabetes. The remaining 1% had other types of diabetes. Type 1 diabetes is usually diagnosed in childhood and insulin replacement is required to survive. In children aged 0–14 the incidence of Type 1 diabetes increased from 19 cases per 100,000 children in 2000 to 25 per 100,000 children in 2004. But since 2004 the rate of new cases has remained stable.

Non-prescription pain relievers

By Madeline Thompson

This education module is independently researched and compiled by PSA-commissioned authors and peer reviewed.

Pain relief is one of the most common reasons people visit a pharmacy. Pain can have a significant impact on a person's quality of life including their physical and mental health, and ability to work, sleep, exercise and perform everyday tasks. Pharmacists can play an important role in assisting people to select an appropriate medicine to treat their pain. Pharmacists can also provide advice on the safe and correct use of pain relieving medicines and pain self management strategies; and recognise when referral to a medical practitioner or other healthcare professional is appropriate.

Learning objectives

After reading this article, pharmacists should be able to:

- Describe the differences between acute and chronic pain.
- Discuss potential side effects and important safety precautions for non-prescription pain relievers.
- Advise on the safe and correct use of non-prescription pain relievers.

Competencies addressed (2010): 1.3, 4.2, 6.1, 6.2, 6.3, 7.2

Severe acute pain, nerve injury and psychosocial factors can predispose to the development of chronic pain.

What is pain?

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Acute pain

Acute pain refers to pain that occurs suddenly and has an identifiable cause associated with injury or tissue damage. It usually has a short duration and is self-limiting. The pain subsides as the injured tissue heals.[1] Common causes of acute pain include: [2]

- muscular pain from sprains, strains or other sports injuries
- back pain
- headache and migraine
- dysmenorrhoea (period pain)
- toothache.

Acute pain can be recurrent or cyclical and have a similar presentation to chronic pain. Conditions that often cause recurrent episodes of acute pain include headaches and dysmenorrhoea.[3]

Chronic pain

Chronic pain is a more complex process and is considered a disease in its own right. Pain is classified as chronic (or persistent) when it persists for more than three months and continues after normal healing.[1] Chronic pain often begins as acute pain. Types of acute pain that may progress to chronic pain include post-operative and post-traumatic pain, acute back pain and shingles pain. As acute pain progresses to chronic pain, nerves in the central nervous system become sensitised and respond in an excessive way to potentially harmful (noxious) stimuli and non-noxious stimuli such as touch and light pressure.[3] Severe acute pain, nerve injury and psychosocial factors can predispose to the development of chronic pain.[2,3] Failure to treat acute pain promptly and appropriately at the time of injury can contribute to the development of chronic pain.

Neuropathic pain is a type of pain associated with injury or dysfunction in the nervous system. It is often described as sharp, stinging or stabbing pain, burning or tingling sensations, or sensitivity to cold or touch.[4] It is a common cause of chronic pain symptoms, but can present acutely following trauma or surgery. Around 3% of people with acute pain experience neuropathic pain.[2] Common causes of neuropathic pain include peripheral nerve damage associated with uncontrolled diabetes, and postherpetic neuralgia due to inflammation of nerve tissues caused by herpes zoster (shingles).[4]

Assessment of pain

Pain assessment tools can be used to understand the type and severity of pain a person is experiencing. For pharmacists, these can help to determine the likely cause of the pain, and whether referral is necessary.

A range of pain assessment tools are available. Multidimensional tools assess several aspects of pain. An example is the "L-I-N-D-O-C-A-R-F-F" mnemonic tool which can be used in a pharmacy setting to obtain a more complete picture of the pain a person is experiencing (see Practice Point 1). Single dimensional assessment tools include numeric, verbal or visual scales that rate only one aspect of pain.[1] With numerical scales, people are asked to use numbers from 0 to 10 to rate the intensity of their pain (0 being no pain and 10 being unbearable pain). Verbal scales contain commonly used words such as 'mild,' 'moderate' and 'severe' to help people describe the severity of their pain (see Table 1. in Counter Connection).

Visual scales use aids such as pictures of facial expressions to illustrate the severity of their pain. An example is the 'Wong Baker Faces Pain Rating Scale'[5] which uses six different facial expressions from happy (no pain) to agony (the worst pain) to indicate how much pain a person feels (see Table 1). Body diagrams may also be used to help pinpoint where the pain occurs.[1] Simple assessment tools can be useful for assessing pain in people with barriers to communication including children, the elderly, and people from a non-English speaking background (see Table 1).

Pain is multifactorial and is influenced by psychological and environmental factors including a person's previous experiences of pain, and their beliefs, mood and coping ability. The perception and experience of pain is subjective and varies between individuals. There is also some individual variation in response to pain-relieving treatments.

Management of pain

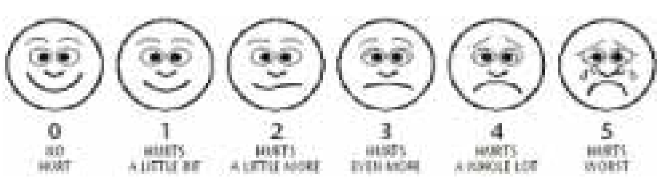
Management of acute pain is directed at symptom relief and treatment of the underlying disorder where possible.[1,2] Acute pain is expected to be of limited duration and resolve as the injured tissue heals. Non-prescription analgesics may be used 'as needed' for relief of mild to moderate pain and to improve mobility until the pain resolves. Neuropathic pain is less likely to respond to conventional analgesics (e.g. paracetamol, NSAIDs and opioids).[4]

Table 1. Simple pain assessment tools[5]

Numeric Rating Scale

0	1	2	3	4	5	6	7	8	9	10
No pain										Unbearable/ worst pain ever)

Wong-Baker FACES Pain Rating Scale



From Hockenberry MJ, Wilson D, Winkelstein ML: Wong's Essentials of Pediatric Nursing, ed. 7, St. Louis, 2005, page 1259. Used with permission. Copyright, Mosby.

Practice point 1

The 'LINDOCARRF' mnemonic for describing aspects of pain:

L – Location(s)
(Where do you feel the pain?)

I – Intensity
(Is is mild, moderate, or severe?)

N – Nature
(e.g. throbbing, sharp, dull, burning)

D – Duration
(How long does the pain last?)

O – Occurrence
(When does the pain occur?)

C – Concurrence
(Other associated symptoms? e.g. nausea, numbness or weakness)

A – Aggravating factors
(What makes the pain worse?)

R – Relieving factors
(What makes the pain better?)

R – Radiating
(Does the pain travel to other body parts?)

F – Frequency
(How often does the pain occur? Is it constant or intermittent?)

Practice point 2

Codeine-containing non-prescription pain relievers [4,6,24]

Codeine is available in combination with paracetamol, aspirin or ibuprofen as *Pharmacist Only* medicine for the temporary relief of moderate to severe acute pain. It is uncertain whether codeine in the amounts contained in these preparations (up to 15 mg per dosage unit) is sufficient to provide additional pain relief compared to paracetamol or NSAIDs alone. It is generally accepted that doses less than 30 mg codeine are subtherapeutic. In addition, approximately 7%–10% of Caucasians and 1%–2% of Asians are 'poor metabolisers' of codeine to morphine (codeine's active metabolite) and obtain negligible analgesic benefit from codeine. These individuals may require alternative analgesic therapy. Codeine has a limited role in chronic pain due to the short duration of action of both codeine and morphine (approximately three hours). Regular use of codeine is commonly associated with drowsiness and constipation. Tolerance, and physical or psychological dependence may also develop with prolonged use. Dependency is more likely in patients with a history of drug or alcohol addiction. Pharmacists should be aware of signs of addictive behavior including repeated requests for codeine-containing analgesics (e.g. because the product has been lost) and attempts to obtain products from multiple sources.

Chronic pain can be caused by a wide variety of conditions and requires clinical assessment and management by a doctor. The psychological factors contributing to the person's emotional and physical experience of pain also need to be considered.

Non-prescription pain relievers

Pain relieving medicines available without a prescription include paracetamol and NSAIDs. These medicines are widely used by consumers, often without the knowledge and advice of a health care professional. Pharmacists can play an important role in advising consumers on the safe and appropriate use of non-prescription analgesics for the management of acute pain, as well as recognising when referral to a doctor or other health care professional is necessary.

Paracetamol and NSAIDs are suitable for short term relief of mild to moderate pain. The most appropriate analgesic for an individual depends on the likely cause of the pain, the presence of risk factors for NSAID-related adverse effects and the relative risk of adverse effects with different NSAIDs. Non-prescription analgesics are frequently used for symptomatic relief of acute pain associated with the following conditions:[2]

- postoperative pain
- musculoskeletal pain (e.g. shoulder, neck or back pain)
- primary dysmenorrhea
- headache and migraine
- cold and flu symptoms (e.g. sore throat, sinus pain)
- dental pain.

Combination analgesics containing codeine may be recommended for temporary relief of moderate to severe acute pain (see Practice Point 2, and PSA's *Guidance document on the 'Provision of a combination analgesic containing codeine as a* Pharmacist Only *medicine'* for more information). Chronic pain may also be managed with certain non-prescription medicines under the supervision of a doctor, with regular review to ensure pain is being adequately controlled.

Paracetamol

Paracetamol is an effective analgesic for mild to moderate pain. The incidence of adverse effects at recommended doses is comparable to placebo.[2] It can be used when NSAIDs are contraindicated or pose an unacceptable risk of side effects.

The recommended dosage in adults and children over 12 years of age is 1–2 tablets (500–1000 mg) every 4–6 hours, up to a maximum of 4 g (8 tablets) per day. If using controlled-release tablets, the dose is 2 tablets (1330 mg) every 6–8 hours,

Paracetamol and NSAIDs are suitable for short term relief of mild to moderate pain.

up to a maximum of six tablets per day. For children over one month of age, dose calculations are based on weight: 15 mg/kg every 4–6 hours up to a maximum of 60 mg/kg per day.[6] Higher doses up to 90 mg/kg/24 hours may be prescribed for up to 48 hours under medical supervision.[7] Paracetamol has been shown to have a dose-response effect, with doses of 1000 mg found to be superior to 500 mg in adults.[2] Paracetamol is also an effective adjunct to opioid analgesia. Opioid requirements are reduced by 20–30% when combined with regular doses of paracetamol.[2]

Safety precautions

Regular dosing for greater than 24–48 hours is not recommended in children and adolescents without medical review. There is an increased risk of paracetamol toxicity in children and adolescents with fever or poor oral intake, even at doses close to therapeutic doses, when given for more than 48 hours.[8]

The routine use of prophylactic paracetamol at the time of infant vaccination is no longer recommended, as it has been associated with reduced antibody response. Paracetamol (or other analgesics or antipyretics) should only be used if post-vaccine symptoms develop.[9,10]

In adults, repeated ingestion of supratherapeutic (excessive) doses can cause liver injury. When advising on the safe and correct use of paracetamol, it is important to clarify whether people are taking other products containing paracetamol and the total daily dose should be calculated where necessary. Paracetamol is involved in a number of cases of poisoning requiring hospitalisation due to inadvertent, repeated supratherapeutic dosing. The risk of liver injury in overdose is increased by chronic alcohol abuse, hepatic impairment (e.g. viral hepatitis, alcoholic hepatitis) concurrent use of some microsomal-inducing drugs (e.g. barbiturates, carbamazepine, rifampicin, isoniazid) and glutathione depletion (e.g. recent prolonged fasting, acute illness with prolonged vomiting or dehydration, anorexia nervosa, bulimia, malnutrition, malignancy, HIV-AIDS).[8] Therapeutic doses of paracetamol are unlikely to cause hepatotoxicity in patients who ingest moderate to large amounts of alcohol.[11] In patients with chronic alcoholism or liver disease, paracetamol may be preferred to NSAIDs but should only be used under medical supervision. Careful dosage instructions are necessary to minimise the potential for accidental overdose.

Regular doses of paracetamol for prolonged periods can increase the anticoagulant effect of warfarin. Increased monitoring and dose adjustment of warfarin may be necessary, particularly for patients who are susceptible to changes in their INR (e.g. the elderly), and those at increased risk or with a history of significant bleeding. Occasional doses of paracetamol (i.e. less than 2 g/day for one day) are generally considered safe in people taking warfarin, with no additional monitoring required. However, regular or higher doses of paracetamol should only be used on medical advice.[12]

Non-steroidal anti-inflammatory drugs (NSAIDs)

Non-prescription NSAIDs are suitable for temporary treatment of mild to moderate

Table 2. Non-prescription NSAIDs [6]

Drug	Recommended doses for non-prescription use
Aspirin	Adults and children over 16 year of age: 300–900 mg every 4–6 hours (maximum 3600 mg per day).
Diclofenac	Adults and children over 14 years of age: 25 mg initially, then 12.5–25 mg every 4–6 hours (maximum of 75 mg per day). Pharmacists may recommend higher doses for short term treatment (up to one week) of acute pain; consult product information for further information.
Ibuprofen	Adults and children over 12 years of age: 200–400 mg every four hours (maximum of 1200 mg per day). Children from 3 months to 12 years of age: the dose is based on the child's weight, 5–10 mg/kg every 6–8 hours (maximum of 3 doses in 24 hours).
Mefenamic acid	Adults and children over 14 years of age: 500 mg three times a day from the onset of period pain for a maximum of seven days.
Naproxen sodium	Adults: 550 mg initially, then 275 mg every 6–8 hours (maximum of 1375 mg per day).

Practice point 3

Some contraindications and precautions with non-prescription NSAIDs that require referral:[4,6]

- Pregnancy
- Other medical conditions
  - Ischaemic heart disease, hypertension, heart failure
  - History of stroke or myocardial infarction
  - Renal or hepatic impairment
  - History of allergy to aspirin / NSAIDs (i.e. asthma, angioedema, urticaria)
  - Inflammatory bowel disease
- Over 65 years of age
- Planned surgery or dental surgery
- Poor response
- Prolonged use for more than 10 days.

Practice point 4

Potential drug interactions with NSAIDs:[6]

- Angiotensin converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARIIB) + diuretic
- Other antihypertensives
- Anticoagulant / antiplatelet therapy (including low dose aspirin)
- Bisphosphonates
- Corticosteroids
- Lithium
- Selective serotonin reuptake inhibitors (SSRIs) / Selective Noradrenaline Reuptake Inhibitors.

Related Fact Cards

» *Back Pain*
» *Colds and Flu*
» *First Aid in The Home*
» *Gout*
» *Headache*
» *Migraine*
» *Opioids for Pain Relief*
» *Osteoarthritis*
» *Pain Relievers*
» *Period Problems*
» *Sprains and Strains*

acute pain, especially where there is inflammation associated with tissue injury. NSAIDs may be the first choice for certain painful conditions such as dysmenorrhoea due to their effect on peripheral prostaglandin synthesis. The minimum effective dose should always be used for the shortest duration necessary to control symptoms. NSAIDs given in addition to paracetamol can improve analgesia compared with paracetamol alone.[2]

Safety precautions

All NSAIDs, including those which can be purchased without a prescription, belong to the same therapeutic class and have similar efficacy. However, at equipotent doses there is some variability in individual response to the effects of different NSAIDs in terms of clinical efficacy and adverse effects. There are some known differences between individual NSAIDs with respect to their anti-inflammatory effects and toxicity. These differences may influence the choice of NSAID used.

Some potential adverse effects and important safety precautions are common to all non-prescription NSAIDs including effects on platelet aggregation and potential to prolong bleeding time. People taking NSAIDs should be advised to seek medical advice before planned surgery. There are important drug interactions with antithrombotic (anticoagulant and antiplatelet) therapies. All NSAIDs have the potential to worsen asthma in people with NSAID-sensitive asthma. NSAIDs trigger asthma symptoms in up to 11% of adults and 2% of children with asthma. Symptoms tend to occur within 1–3 hours of taking a dose, and include runny nose, shortness of breath and wheezing. NSAIDs are not contraindicated if a person with asthma has taken NSAIDs previously without any worsening of their asthma symptoms.[13] NSAIDs should also be avoided in pregnancy. See Practice Point 3 for a list of common precautions and contraindications to the use of non-prescription NSAIDs.

The gastrointestinal effects of NSAIDs vary from dyspepsia, heartburn, and abdominal discomfort to more serious ulceration with potentially life-threatening bleeding and perforation. Serious gastrointestinal bleeding or perforation is rare and mainly occurs in people with risk factors for NSAID-related gastrointestinal bleeding.

The presence of multiple risk factors greatly increases the risk of gastrointestinal complications.[14] People with any of the following risk factors for gastrointestinal bleeding should be advised to avoid using NSAIDs without medical advice:[14,15]

- Age $\geq$ 65 years
- History of peptic ulcer disease
- *Helicobacter pylori* infection
- Multiple NSAIDs
- Prolonged use of NSAIDs at maximal doses
- Anticoagulant or antiplatelet therapy (including low-dose aspirin)
- Oral corticosteroids.

The presence of serious co-morbidity including cardiovascular disease, renal or hepatic impairment, diabetes or hypertension, is an additional risk factor for NSAID-related gastrointestinal complications.[15]

Ulcer complications are often asymptomatic. Conversely, dyspepsia and other gastrointestinal symptoms are not always a reliable indicator of mucosal damage.[15] However, people with signs and symptoms suggestive of NSAID-related gastroesophageal irritation require referral. These include:[14,15]

- Dyspepsia
- Heartburn
- Bloating or cramping
- Nausea and vomiting
- Retrosternal or substernal chest pain
- Difficulty or pain on swallowing
- Vomiting blood or blood in faeces
- Unexplained weight loss.

Comparative information

The relative gastrointestinal and cardiovascular safety of the non-prescription NSAIDs remains uncertain.

Gastrointestinal safety

In healthy individuals with no major risk factors for gastrointestinal bleeding, non-prescription NSAIDs are well tolerated when used short term for up to 10 days at approved doses.[15–17] Ibuprofen is the most widely used over-the-counter NSAID. In healthy individuals without risk factors, one study found ibuprofen (up to 1.2 g /day) to have similar gastrointestinal tolerability to paracetamol (up to 3 g /day), and better tolerability than aspirin (up to 3 g/day) when used short term for up to

seven consecutive days. Tolerability was evaluated according to the frequency of gastrointestinal adverse events including dyspepsia, nausea, diarrhoea, and abdominal pain.[16] Based on prescription data, ibuprofen (at doses up to 1.2 g/day) has the lowest risk of gastrointestinal complications while diclofenac and naproxen have intermediate risks.[15,17] However, information on the comparative rates of gastrointestinal adverse events with NSAIDs at approved non-prescription doses is limited.

The gastrointestinal tolerability of ibuprofen and other NSAIDs is dose dependent and the risk of gastrointestinal bleeding increases in a dose-dependent manner.[15,19,20] Research has found that people often exceed the recommended doses of analgesics and take more in an attempt to obtain quicker or stronger pain relief.[18] It is important that people are warned not to exceed the maximum recommended dose and to limit their use of non-prescription NSAID to no more than 10 days duration without advice from a doctor.

The risk of gastrointestinal adverse effects is known to vary between individual NSAIDs and appears to be influenced by several factors: [15,16,20,21]

- the dose – the risk of gastrointestinal complications increases at higher doses
- duration of treatment – treatment with NSAIDs for more than 1–3 months can cause damage to the gastrointestinal mucosa, especially in people with risk factors for gastrointestinal effects
- the half-life– NSAIDs with a long half-life (or slow-release formulations) are associated with an increased risk of gastrointestinal complications compared to those with a short half-life
- physiochemical properties of the agent (see Practice Point 6 for more information).

When low-dose aspirin is taken in combination with other NSAIDs, the risk of developing serious gastrointestinal complications increases significantly.[14,15]

Cardiovascular safety

There is limited information on the cardiovascular safety of NSAIDs at recommended doses for non-prescription use. Most studies have examined the comparative risks with NSAIDs at typical prescription doses.

A 2011 study found the use of prescribed NSAIDs to be associated with an increased risk of death and recurrent myocardial infarction (MI) in patients with a history of MI. The risk of cardiovascular events was independent of treatment duration, increasing almost immediately with diclofenac and COX-2 inhibitors. The contribution of other cardiovascular risk factors (e.g. blood pressure, smoking, obesity) and the concurrent use of non-prescription NSAIDs such as ibuprofen were not examined.[22] NSAID use has also been found to increase cardiovascular risk in healthy individuals. Naproxen and low-dose ibuprofen are thought to have a lower cardiovascular risk than other NSAIDs.[23] There appears to be a dose-response relationship and more research is necessary to examine the effects of dose, treatment duration, and patient characteristics on the risk of cardiovascular events.[22,23] More research is also needed to assess the comparative safety of different non-prescription NSAIDs, which are generally used at lower doses and for short durations.

NSAIDs should be avoided or used with caution in established cardiovascular disease (e.g. previous MI or heart failure). If NSAID therapy is considered necessary, treatment should be limited to the shortest duration possible. Long term use of

Practice point 5

Relative efficacy of non-prescription analgesics [17,26]

There is limited information available on the equipotent doses of non-prescription pain relievers. The Oxford league table can be used to compare the relative efficacy of different analgesics in acute pain. Analgesic efficacy is expressed as the Number Needed to Treat (NNT), the number of patients who need to receive a drug for one patient to achieve at least 50% relief of moderate to severe acute pain over a 4–6 hour treatment period compared with placebo. The table reflects the dose response to different doses of non-prescription drugs with the most effective drugs having a low NNT of just over 2. For example, ibuprofen 400 mg is approximately equivalent diclofenac 50 mg for relief of moderate to severe pain. When interpreting the information from the table, it is important to note that treatment efficacy varies with different pain symptoms. For example, NSAIDs are more effective for relief of acute pain associated with primary dysmenorrhea compared to paracetamol.

Table. 4 Information from the 'Oxford league table of analgesic efficacy'[26]

Analgesic	NNT
Diclofenac 50 mg	2.3
Ibuprofen 400 mg	2.4
Ibuprofen 200 mg	2.7
Diclofenac 25 mg	2.8
Naproxen sodium 550 mg	3.0
Paracetamol 1000 mg	3.8

Practice point 6

Gastrointestinal adverse effects of NSAIDs[14,15]

The gastrointestinal adverse effects of NSAIDs are thought to be mediated by two different mechanisms:

- local injury to the gastrointestinal tract mucosa from direct contact between the drug and mucosa; and
- systemic inhibition of COX enzyme which reduces the cytoprotective effects of prostaglandins.

The relative gastrointestinal safety of individual NSAIDs is thought to be due to differences in inhibition of different isoforms of COX enzyme, particularly COX-1 and COX-2. COX-1 mediates the formation of prostaglandins which are involved in gastrointestinal mucosal protection. Inhibition of these prostaglandins results in decreased synthesis of gastric mucus and bicarbonate, impaired mucosal blood flow, and increased acid secretion. COX-2 catalyses the synthesis of prostaglandins that act as inflammatory mediators in response to tissue injury. COX-2 selective NSAIDs have an improved upper gastrointestinal safety profile.

The potential for NSAIDs to cause gastrointestinal injury is also dependent on their physicochemical properties including their acidic properties (as measured by pKa). These properties determine their topical irritancy in the gastrointestinal mucosa as well as their systemic bioavailability. Gastric injury has been found to correlate significantly with the pKa of the compound: the lower the acidity of the drug, the less the mucosal damage. For example, the selective COX-2 inhibitor celecoxib is non-acidic with a high pKa value. This may account for the low rate of mucosal irritancy, separate from its low inhibitory activity on COX-1 in the stomach. Most non-selective NSAIDs are organic acids with a low pKa (2.8–4.4) which interact with phospholipids causing disruption to the gastric mucosal barrier. Ibuprofen has a pKa of 5.2, while aspirin, indomethacin and naproxen have pKa values ranging from 3.5 to 4.2. This may help to explain the differences in gastrointestinal tolerability and ulcerogenic activity between these NSAIDs.

NSAIDs with low dose aspirin may reduce the cardioprotective benefits of low dose aspirin.[17] This combination should only be used on medical advice.

Other treatment options for management of pain

Other options for management of acute pain include heat packs, topical creams and ointments, rest, and physical therapy.

Topical NSAIDs [2,6]

Available topical NSAIDs include benzydamine, diclofenac, ibuprofen, ketoprofen and piroxicam. These are available as topical creams/gels for application to the skin. They are normally massaged gently into the affected areas two to four times a day for the short term treatment (1–2 weeks) of musculoskeletal inflammation and injury such as sports injuries, sprains, tendonitis and swelling.

Complementary and alternative medicine (CAM) therapies

CAM therapies are becoming increasing popular for self-treatment of common health problems including pain. These therapies may be used on their own or in addition to conventional medicines and they are often used without the knowledge or advice of health care professionals. This increases the risk of side effects and drug interactions. Effective communication between pharmacists and consumers about their self-treatment choices is important to ensure the use of CAM therapies is safe and appropriate. Herbal products and supplements commonly used for self-treatment of pain include arnica, devils claw, evening primrose oil, fish oil/omega-3 oils, garlic, glucosamine +/- chondroitin, MSM, S-adenosylmethionine (SAMe) and willow bark.[24]

References

1. American Pain Foundation. A Primer on Pain and Its Management. A Policymaker's guide to understanding pain & its management [online]. At: www.painfoundation.org/learn/publications/files/PainMgmtPrimer.pdf
2. Australian and New Zealand College of Anaesthetists and Faculty of Pain Medicine. Acute pain management: scientific evidence. 3rd edn [online] 2011. At: www.anzca.edu.au/resources/college-publications/Acute%20Pain%20Management/books-and-publications/acutepain.pdf
3. National Pain Summit initiative. National Pain Strategy [online]. 2011. At: http://www.anzspm.org.au/c/anzspm?a=sendfile&ft=p&fid=1320268502&sid=
4. eTG complete [internet]. Melbourne: Therapeutic Guidelines; 2011.
5. American Pain Foundation. Target Chronic Pain Notebook [online]. 2008. At: www.painfoundation.org/learn/publications/files/TargetNotebook.pdf
6. Product information. eMIMS [CD-ROM]. Sydney, Australia: CMPMedica Australia; 2011.
7. Kemp CA, McDowell JM, eds. Paediatric Pharmacopeia. 13th edn. Melbourne: Pharmacy Department, Royal Children's Hospital; 2005.
8. Murray L, Daly F, McCoubrie D, et al. Toxicology Handbook. 2nd edn. Sydney: Churchill Livingstone; 2011.
9. Prymula R, Siegrist C, Chlibek R, et al. Effect of prophylactic paracetamol administration at time of vaccination on febrile reactions and antibody responses in children: two open-label, randomized controlled trials. The Lancet. 2009;374(9698):1339–50.
10. Vaccination Procedures. Adverse events following immunisation [revised Jul 2009]. In: The Australian Immunisation Handbook, 9th edn [online]. 2008. At: www.immunise.health.gov.au
11. Graham GG, Scott KF, O'Day R. Alcohol and paracetamol. Aust Prescr. 2004;227:14–5.
12. Hughes GJ, Patel PN, Saxena N. Effect of acetaminophen on international normalized ratio in patients receiving warfarin therapy. Pharmacotherapy. 2011;31(6):591–97.
13. National Asthma Council Australia. Aspirin/NSAID-intolerant asthma: pharmacy notes [online]. 2009. At: www.nationalasthma.org.au
14. Scarpignato C, Hunt RH. Nonsteroidal antiinflammatory drug-related injury to the gastrointestinal tract: clinical picture, pathogenesis, and prevention. Gastroenterol Clin N Am. 2010; 39:433–64.
15. Gastroenterology Society of Australia. NSAIDs and the Gastrointestinal Tract 3rd edn [online]. 2009. At: www.gesa.org.au/files/editor_upload/File/DocumentLibrary/Professional/NSAIDS_Clinical.pdf
16. Moore N, Van Ganse E, Le Parc J, et al. The PAIN study: paracetamol, aspirin and ibuprofen new tolerability study. Clinical Drug Investigation. 1999;18(2):89–9814.
17. Ong SKS, Lirk P, Tan CH, et al. An evidence-based update on nonsteroidal anti-inflammatory drugs. Clinical Medical Research. 2007;5(1):19–34.
18. Biskupiak JE, Brixner DI, Howard K, et al. Gastrointestinal complications of over-the-counter non-steroidal anti-inflammatory drugs. Journal of Pain & Palliative Care Pharmacotherapy. 2006;(20)3:7–14
19. Masso Gonzalez EL. Patrignani P, Tacconelli S, et al. Variability among nonsteroidal antiinflammatory drugs in risk of upper gastrointestinal bleeding. Arthritis & Rheumatism. 2010;62(6):1592–1601.
20. Smecuol E, Bai JC, Sugai E. Acute gastrointestinal permeability responses to different non-steroidal anti-inflammatory drugs. Gut [online]. 2001;49:650–55. At: http://gut.bmj.com/content/49/5/650.full.pdf+html
21. Rodriguez LAG, Hernandez-Diaz S. Relative risk of upper gastrointestinal complications among users of acetaminophen and nonsteroidal anti-inflammatory drugs. Epidemiology [online]. 2001;12(5):570–76. At: http://213.4.18.135:48080/68.pdf
22. Schjerning Olsen A, Fosbol EL, Lindhardsen J, et al. Duration of treatment with non-steroidal anti-inflammatory drugs and impact on risk of death and recurrent myocardial infarction in patients with prior myocardial infarction – A nationwide cohort study. Circulation [online]. 2011. At: http://circ.ahajournals.org/content/123/20/2226
23. McGettigan P, Henry D. Cardiovascular risk with non-steroidal anti-inflammatory drugs: systematic review of population-based controlled observation studies. PLoS Med [online] 2011;8(9): e1001098. doi:10.1371/journal.pmed.1001098
24. Vallerand AH, Fouladbakhsh JM, Templin T. The use of complementary/alternative medicine therapies for the self-treatment of pain among residents of urban, suburban, and rural communities. American Journal of Public Health [online]. 2003; 93(6): 923–25. At: www.ncbi.nlm.nih.gov/pmc/articles/PMC1447871/
25. Sansom L, ed. Australian Pharmaceutical Formulary and Handbook. 22nd edn. Canberra: Pharmaceutical Society of Australia; 2012.
26. Oxford league table of analgesics in acute pain. Bandolier [online]. 2007. At: www.medicine.ox.ac.uk/bandolier/booth/painpag/acutrev/analgesics/leagtab.html

Assessment questions for the pharmacist

Select one correct answer from each of the following questions.
Answers due 31 May 2012.

Before undertaking this assessment, you need to have read the Facts Behind the Fact Card article in *inPHARMation*, and the associated Fact Cards.This activity has been accredited by PSA as a Group 2 activity. Two CPD credits (Group 2) will be awarded to pharmacists with eight out of 10 questions correct. PSA is authorised by the Australian Pharmacy Council to accredit providers of CPD activities for pharmacists that may be used as supporting evidence of continuing competence.

Submit answers online

To submit your response to these questions online, go to the PSA website **www.psa.org.au/selfcare**

AUSTRALIAN PHARMACY COUNCIL ACCREDITED CPD

UP TO 2 CPD CREDITS GROUP 2

Accreditation number: CS120003

This activity has been accredited for Group 2 CPD (or 2 CPD credits) suitable for inclusion in an individual pharmacist's CPD plan.

1. Which of the following are risk factors for developing chronic pain?

a. Poor management of acute pain.
b. Severe acute pain.
c. Nerve injury.
d. All of the above.

2. Choose the CORRECT statement about paracetamol.

a. Less than 3 g/day for one day is generally considered safe in people taking warfarin.
b. Paracetamol is contraindicated in people who ingest moderate to large amounts of alcohol.
c. Paracetamol is generally safe in overdose.
d. Routine use of paracetamol at the time of infant vaccination is no longer recommended.

3. Choose the INCORRECT statement about NSAIDs.

a. There is improved pain relief when NSAIDs and paracetamol are taken together.
b. NSAID are contraindicated in people with a history of asthma.
c. All non-prescription NSAIDs can affect platelet aggregation and prolong bleeding time.
d. NSAIDs should be used with caution in people taking corticosteroids.

4. NSAIDs should only be used on medical advice in:

a. Ischaemic heart disease.
b. Patents with a history of stroke.
c. Inflammatory bowel disease.
d. Planned surgery or dental surgery.
e. All of the above.

5. Choose the INCORRECT statement about NSAID-related gastrointestinal toxicity.

a. When low-dose aspirin is taken in combination with NSAIDs the risk of serious gastrointestinal events is decreased.
b. There is a low risk of gastrointestinal complications when non-prescription NSAIDs are used short term for up to 10 days at approved doses.
c. The gastrointestinal tolerability of NSAIDs is dose dependent and the risk of gastrointestinal bleeding increases in a dose-dependent manner.
d. Treatment with NSAIDs for more than 1–3 months can cause damage to the gastrointestinal mucosa, especially in people with risk factors for gastrointestinal effects.

6. Which of the following is a possible sign of NSAID-related gastroesophageal irritation?

a. Difficulty or pain on swallowing.
b. Retrosternal or substernal chest pain.
c. Dyspepsia.
d. Bloating or cramping.
e. All of the above.

7. Choose the INCORRECT statement about NSAID-related cardiovascular risk.

a. NSAIDs should be avoided or used with caution in established cardiovascular disease.
b. Long term use of NSAIDs in combination with low dose aspirin may improve the cardioprotective benefits of low dose aspirin.
c. The risk of cardiovascular events associated with NSAID use is thought to be dose dependent.
d. NSAIDs increase the risk of recurrent myocardial infarction (MI) in patients with a history of MI.

8. Which of the following medicines should be used with caution in combination with NSAIDs?

a. Venlafaxine.
b. Low dose aspirin.
c. Bisphosphonates.
d. Lithium.
e. All of the above.

9. Choose the INCORRECT statement about gastrointestinal toxicity caused by NSAIDs.

a. Local injury to the gastrointestinal tract mucosa contributes to the gastrointestinal adverse effects of NSAIDs.
b. Systemic inhibition of cyclooxygenase (COX-1) enzyme by NSAIDs reduces the cytoprotective effects of prostaglandins.
c. NSAIDs with low pKa values (2.8–4.4) have lower rates of gastrointestinal irritancy.
d. Prostaglandins stimulate mucus and bicarbonate production and decrease acid secretion in the gastrointestinal tract.

10. Which of the following factors influences the risk of gastrointestinal adverse effects with different NSAIDs?

a. Dose.
b. Presence of risk factors.
c. Treatment duration.
d. Half-life.
e. All of the above.

Be prepared for Heart Week in May

By Deborah Cracknell, Manager Practice Program

Self care and the National Heart Foundation have joined forces on the May issue of *inPHARMation* in the lead up to Heart Week.

The Heart Foundation is providing several wallet-sized cards for consumers which will be included in your May *inPHARMation*.

It will also include a new Fact Card on Heart Attack. Of course, the Facts Behind the Fact Card and Counter Connection articles will focus on heart attack so that pharmacy staff have access to the latest information.

We are distributing the May issue of *inPHARMation* a week early so Self Care members can use the resources to participate in Heart Week which runs from Sunday 6 May to Saturday 12 May.

This campaign points out that no two heart attacks are the same and that someone who has already had a heart attack may have different symptoms the second time. Knowing the warning signs of heart attack and acting quickly can reduce the damage to your heart muscle and increase your chance of survival.

According to the National Heart Foundation, an Australian dies of a heart attack every 51 minutes and more than 50% of people who die of a heart attack don't make it to hospital. Too many people delay calling triple-0 because they:

- tend to ignore their warning signs or wait for them to go away
- don't recognise their symptoms as warning signs of heart attack
- think their symptoms need to be severe to call an ambulance
- don't want to bother the ambulance service or don't want other people to worry about them
- think it's better to call their GP.

These are important messages that Self Care pharmacies can highlight during Heart Week.

Please use the resources provided in the May *inPHARMation* to support the National Heart Foundation to spread the message.

Self Care at APP

In March PSA attended APP on the Gold Coast. Self Care Project Coordinator, Kylie Davis and other PSA staff spent the weekend promoting the benefits on Self Care membership, meeting current members and recruiting new members.

As a result Self Care has four new members.

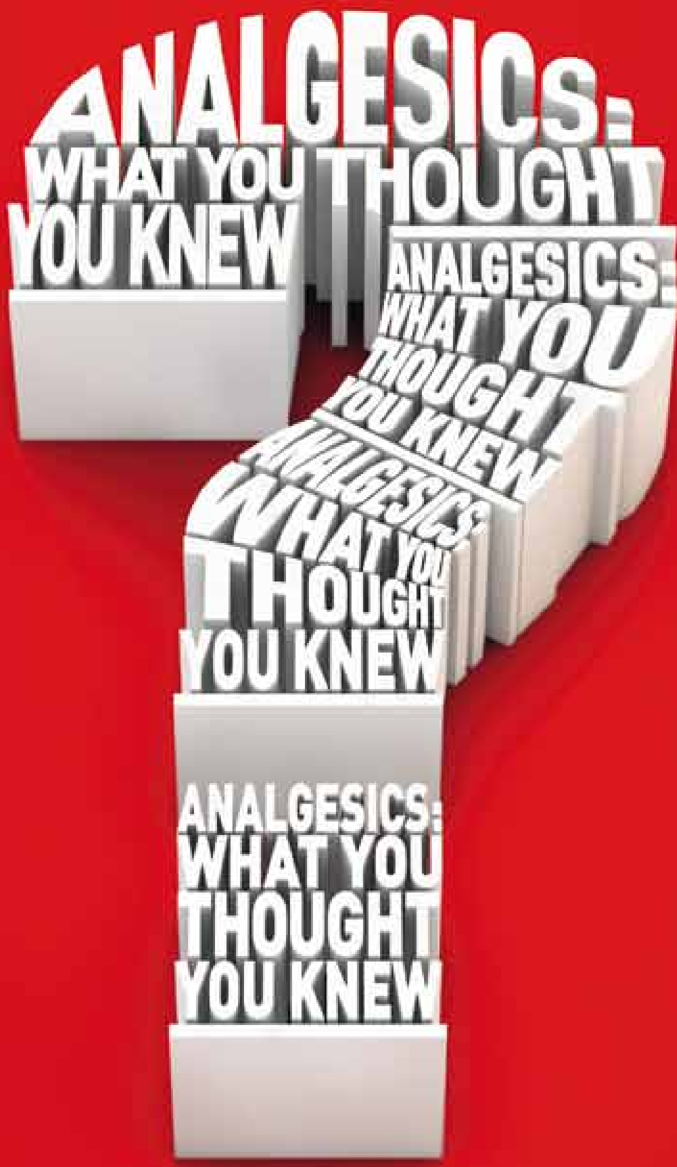
We would like to welcome all of our new members from March.

New Self Care Members

Southside Pharmacy Mudgee – NSW
Health Information Pharmacy Morningside – QLD
Health Information Pharmacy Brimbank – VIC
Dalwallinu Pharmacy – WA
Collinsville Pharmacy – QLD
Goodna Day & Night Pharmacy – QLD
Kings Circle Pharmacy – QLD
Galleon Way Pharmacy – QLD
Pharmacy Select Highlands – VIC
Biloela Discount Chemist – QLD
Fernvale Discount Drugstore – QLD
St Andrews Pharmacy – WA

Dextropropoxyphene

NPS has published information online (at: www.nps.org.au) to help prepare for the TGA's cancellation of Capadex and Paradex, two of the four medicines available in Australia containing dextropropoxyphene. This information outlines recommendations for prescribers, new contraindications and alternatives to dextropropoxyphene for analgesia. NPS CEO Dr Lynn Weekes said that the cancellation reflects concerns that dextropropoxyphene has an unfavourable balance of analgesic efficacy and toxicity. 'While Di-Gesic and Doloxene will remain on the market for the time being, the advice remains the same for all these medicines – the potential harms outweigh the possible benefits,' says Dr Weekes. NPS recommends prescribers should not initiate dextropropoxyphene treatment for any new patients and should review all patients currently taking dextropropoxyphene.



EVERY 15 MINUTES YOUR STAFF COULD BE RECOMMENDING THE WRONG ANALGESIC[1]

Help your staff give the right advice with the new Quality Use of Medicines education kit. *Analgesics: What you thought you knew* will bring your pharmacy up to speed with the current views of GI experts on the tolerability of OTC analgesics.

ORDER YOUR KIT TODAY

Call 1800 226 766 to find out more.

Reference: 1. Metis Market Research, 2011. The kit represents 1 hour of QCPP Approved Refresher Training for pharmacy assistants. Reckitt Benckiser Australia. 44 Wharf Road, West Ryde, NSW 2114. RECB5635/IPH/FP. 03/12.

NUROFEN

The Pharmacy Guild of Australia

Reckitt Benckiser

Non-prescription pain relievers

By Madeline Thompson

This education module is independently researched and compiled by PSA-commissioned authors and peer reviewed.

People often choose to self-treat pain using non-prescription pain relievers. When used short-term, these medicines are safe for most people to take. However, all medicines need to be used carefully to work effectively and avoid possible side effects and drug interactions. As pharmacy assistants, you can play an important role in helping customers to select an appropriate medicine to treat their pain and in advising them on the safe and correct use of pain relievers. You can also assist customers by recognising when it is appropriate to refer to the pharmacist for further advice.

APPROVED REFRESHER TRAINING

Supporting Excellence in Pharmacy

Pain can worsen anxiety and other emotional problems, and interfere with mood, sleep and work.

What is pain?

Pain is the unpleasant sensation or feeling we experience when the body is damaged or stressed. Pain is experienced when nerves carry a message from the affected area of the body to the brain. It can be caused by surgery, injury or medical conditions, although sometimes the cause of the pain cannot be found. The type and severity of the pain will depend on the area of the body affected (e.g. the skin, bones and joints, or internal organs). Pain may be felt in one area of the body, or be widespread. It can be mild or severe and range from a dull ache or cramp to sharp, stabbing or throbbing pain. Everyone responds to pain differently. Emotional stress and fatigue can make pain feel worse, and different people can tolerate different degrees of pain. Pain can worsen anxiety and other emotional problems, and interfere with mood, sleep and work.

Types of pain

Pain is the most common reason people self-treat using medicines available without a prescription from a pharmacy, supermarket or other shop. This often occurs without the knowledge or advice of a health care professional. Advice and information from trained pharmacy staff is important to make sure use of non-prescription pain relievers is safe and appropriate.

Common types of pain that people often try to self-treat include sports injuries (e.g.sprains, strains), back pain, headache, migraine, period pain, osteoarthritis and toothache. Whenever you recommend a pain reliever or offer advice about how best to manage pain, it is important to find out whether the pain is acute or chronic, and the severity of the person's pain. This will help to determine whether it is appropriate for the customer to self-treat using a non-prescription pain reliever or whether referral to the pharmacist is necessary.

Acute and chronic pain

Acute pain starts suddenly and lasts for a short period of time (minutes to weeks). There is usually an obvious cause such as a cut or graze, headache, broken bones or recent surgery. This type of pain tends to go away as the body heals.

Chronic pain (or 'persistent pain') is pain that lasts for three months or longer. This type of pain can be related to medical conditions such as arthritis or cancer. It can also be pain that continues after the original cause has gone. Chronic pain can develop from acute pain that wasn't managed properly. Chronic pain can impact on peoples' ability to carry out normal daily activities, and can affect their emotional and physical health and quality of life. Some people think that chronic pain is something they have to live with. This is not correct as most pain can be relieved with proper pain management.

Chronic pain can be a type of pain called neuropathic pain. Neuropathic pain is caused by damaged nerves that become very sensitive to stimuli such as touch or pressure that wouldn't normally cause pain. Neuropathic pain is often described as tingling, burning, shooting or electric-type pain. Causes of neuropathic pain include nerve damage from an accident or surgery, post-herpetic neuralgia (pain that persists after shingles), uncontrolled diabetes and multiple sclerosis. Effective treatment of neuropathic pain generally requires prescription medicines.

Table 1. Pain scale to help assess pain severity

Pain severity	Scale from 1–10	Description
Minor	1–3	Barely noticeable, uncomfortable, tolerable or bearable. Does not interfere with most activities. Can find ways to adapt to the pain or get used to it.
Moderate	4–6	Strong pain that is noticeable most or all of the time. Cannot find ways of completely adapting to the pain.
Severe	7–10	Unbearable, excruciating or very intense. Interferes with thinking, concentration, mood, appetite, energy levels and sleep.

People who appear to be experiencing severe pain require referral to a pharmacist.

Severity of pain

Mild to moderate acute pain can often be managed with non-prescription pain relievers. However, the severity of the pain a person is experiencing can be difficult to assess. What is mild to one person may be unbearable to another. Sometimes it helps to ask customers to rate their pain on a scale from 1–10, where 1 is little or no pain and 10 is unbearable pain (see Table 1). Severe pain may be described as distressing, intolerable or intense, and may be affecting a person's sleep, mood or concentration. People who appear to be experiencing severe pain require referral to a pharmacist.

Management of pain

Pharmacy assistants may be asked to assist customers to choose a medicine to treat their pain. You can play an important role by recommending an appropriate pain reliever or referring the customer to a pharmacist.

Pain can be managed with pain relieving medicines or other measures which may be used alone or in combination with pain relieving medicines. These include massage, acupuncture, application of heat or cold, complementary medicines, braces and strapping, or exercises recommended by a physiotherapist. Acute pain and chronic pain are managed in very different ways. Mild to moderate acute pain can be managed with non-prescription pain relievers to provide quick, short term pain relief until the cause of the pain goes away. Any customer with stronger acute pain or persistent, chronic pain should be referred to the pharmacist. Non-prescription medicines should only be used long term under medical supervision.

Non-prescription pain relievers

There are two main types of pain relievers available without a prescription: paracetamol and NSAIDs. The decision to recommend a particular pain reliever or refer to the pharmacist will depend on a number of factors including the type, cause, severity, and duration of the pain; whether the person has already tried medicines for pain relief; and whether they are taking other medicines or have other medical conditions.

Paracetamol

Paracetamol is often the first choice of pain relief for mild to moderate pain. Paracetamol is a very safe medicine when used at recommended doses. It has a low risk of side effects and interactions with other medicines.

Various products containing paracetamol are advertised for different types of pain (e.g. *Panadol Back and Neck Pain Relief*). Paracetamol is also available in combination with other ingredients such as phenylephrine and codeine for relief of cold and flu symptoms. Customers need to be aware of all products they are taking that contain paracetamol. Accidental overdose with paracetamol can occur when people use more than one paracetamol product at a time without realising it. Paracetamol is one of the most common causes of accidental overdose in Australia. In larger than recommended doses it can cause liver damage. Any person suspected of taking an overdose (on purpose or accidentally) should be referred to a hospital immediately.

Recommended dosage:

- Adults and children over 12 years of age: one tablet (500 mg) or two tablets (1 g) every 4–6 hours. Maximum of 4 g (8 tablets) per day.
- If using controlled-release tablets: two tablets (1330 mg) every 6–8 hours. Maximum of six tablets per day.
- Children over one month of age: 15 mg/kg every 4–6 hours. Maximum of 60 mg/kg per day.

Customers who have not obtained pain relief from paracetamol may not be taking an adequate dose, or they may need a stronger pain reliever. If customers have been taking paracetamol and continue to experience pain, it is important to ask what dose they have been using. Paracetamol is available in combination with codeine as *Pharmacist Only* medicine for stronger pain relief. People requesting codeine-containing products or treatment for strong pain should be referred to the pharmacist.

Non-steroidal anti-inflammatory medicines (NSAIDs)

NSAIDs available without a prescription include aspirin, diclofenac, ibuprofen, mefenamic acid and naproxen. Most of these medicines are *Pharmacy* medicines and are only available from pharmacies. Small packs of aspirin and ibuprofen can also be purchased from supermarkets and other shops. All NSAIDs work in a similar way and may be used to treat mild to moderate pain. NSAIDs may be preferred to paracetamol for certain types of pain including inflammatory pain (e.g. muscle strains and sprains) and period pain. NSAIDs can cause side effects and appropriate advice from trained pharmacy staff will help to make sure they are used safely and correctly.

All NSAIDs can cause gastrointestinal side effects (e.g. nausea, indigestion, heartburn and discomfort in the upper part of the stomach). In the low doses approved for non-prescription use, ibuprofen is least likely to cause these side effects. Some research involving healthy people with no gastrointestinal conditions has found that when low dose ibuprofen is used for only a few days, it has similar rates of gastrointestinal side effects to paracetamol. However, when used in higher doses or for longer than 7–10 days, the gastrointestinal side effects of ibuprofen and other NSAIDs are more likely to occur. If used long term, NSAIDs can cause ulceration and bleeding in the stomach or intestine. NSAIDs should not be used for self-treatment of pain for more than seven days without medical supervision. Customers who have already been using a NSAID for seven days should be referred to the pharmacist. If people experience an upset stomach from NSAIDs, taking doses with or soon after food can help to reduce gastrointestinal side effects.

To reduce the possibility of side effects, the lowest effective dose should be used for the shortest period of time and only one oral NSAID should be used at a time. Use of more than one oral NSAID can increase the risk of side effects without any further improvement in symptoms. However, sometimes doctors recommend that people take low dose of aspirin (75–150 mg) to prevent blood clots, as well as a NSAID for pain relief. This combination should only be used on doctors' advice.

NSAIDs are not always suitable because of possible side effects or drug interactions. NSAIDs can interact with a number of other medicines. Customers who are taking other medicines should always be referred to the pharmacist. NSAIDs can also interfere with some other medical conditions such as asthma. About 3–11% of people with asthma experience worsening asthma symptoms or an asthma attack after taking NSAIDs. Symptoms tend to occur within 1–3 hours of taking a dose, and include runny nose, shortness of breath and wheezing. If a person with asthma has previously taken NSAIDs without any worsening of their asthma

Table 2. Non-prescription non-steroidal anti-inflammatories (NSAIDs)

	Recommended dosage	Important information
Aspirin	Adults: 300–900 mg every 4–6 hours (maximum of 3600 mg per day).	Avoid in children under 16 years of age unless directed by a doctor. It can increase the risk of Reye's syndrome, which is potentially fatal.
Diclofenac	Adults and children over 14 years of age: 25 mg initially, then 12.5–25 mg every 4–6 hours (maximum of 75 mg per day).	Higher doses may be recommended by a pharmacist or prescribed by a doctor. Available in a 25 mg strength tablet as a S3 *Pharmacist Only* medicine.
Ibuprofen	Adults and children over 12 years of age: 200–400 mg three or four times a day (maximum of 1200 mg per day). Children from 3 months to 12 years of age: the dose is based on the child's weight. Recommended dose is 5–10 mg/kg three to four times a day (maximum of 1200 mg per day).	Some ibuprofen products are advertised for the treatment of specific types of pain, for example Nurofen Tension Headache and Nurofen Migraine Pain. Make sure customers are aware these products contain the same medicine, to make sure they do not double up and increase their risk of side effects. Ibuprofen is the only oral NSAID recommended for use in children under 5 years of age. It can be used in infants from 3 months of age.
Mefenamic acid	Adults and children over 14 years of age: 500 mg three times a day from the onset of period pain for a maximum of seven days.	Marketed for period pain, although all NSAIDs are considered equally effective.
Naproxen sodium	Adults: 550 mg initially, then 275 mg every 6–8 hours (maximum of 1375 mg per day).	

symptoms, then NSAIDs are generally safe to use. NSAIDs can also prolong bleeding time. It may be necessary to avoid taking NSAIDs before surgery because of the increased risk of bleeding. Any customer who requests a NSAID and has other medical conditions or has surgery or dental surgery planned should be referred to the pharmacist.

Other treatments for pain

Topical NSAIDs including benzydamine, diclofenac, ibuprofen, ketoprofen and piroxicam are available as topical creams/ gels for application to the skin. These products are normally massaged gently into the affected areas two to four times a day for the short term treatment (1–2 weeks) of musculoskeletal inflammation and injury such as sports injuries, sprains, tendonitis and swelling. Topical NSAIDs are absorbed through the skin into the blood stream in small amounts and are generally considered safer but less effective than oral NSAIDs. They should be used with caution in people taking certain medicines and with certain medical conditions. Oral and topical NSAIDs should only be used together on medical advice.

Other treatments for pain include acupuncture, massage, transcutaneous electrical nerve stimulation (TENS), hot and cold packs, braces and strapping, complementary medicines and exercises that are tailored by a physiotherapist. Some of these treatments are available from pharmacies.

Related Fact Cards

- *Back Pain*
- *Colds and Flu*
- *First Aid in The Home*
- *Gout*
- *Headache*
- *Migraine*
- *Opioids for Pain Relief*
- *Osteoarthritis*
- *Pain Relievers*
- *Period Problems*
- *Sprains and Strains*

Table 3. Pharmacy assistant protocol for managing symptom-based requests for non-prescription pain relievers
(adapted from May 2008 inPHARMation)

Check	Important information
Who is experiencing the pain?	Children under 12 years of age Pregnant or breastfeeding People aged over 65 years
What are the symptoms?	Pain: • is severe • is getting worse • does not have an obvious cause • reduces ability to function • shows signs of being neuropathic pain • possible side effects
How long have the symptoms been present?	Pain has been present for more than 3 months Pain relievers have been used for more than 7 days without medical advice
What has been tried?	Non-prescription pain relievers have been tried and are ineffective at recommended doses Possible side effects from use of non-prescription pain relievers
Using any other medicines?	Any other medicines
Any other medical conditions?	Any other medical conditions Allergy Planned surgery or dental surgery
Assess	
Is the diagnosis clear?	No obvious cause of pain
Is a non-prescription pain reliever the most appropriate treatment?	Not clear if medicine is appropriate
Are there any possible interactions?	Any other medicines
Are you trained and confident to manage this?	Not trained and/or confident
Respond	
Recommend appropriate therapy	Uncertain
Explain	
Verbal directions *(Dosage, how often, when to stop, other product specific information)*	
Written support *(e.g. Self Care Fact Cards)*	
What to do if no improvement When to see a doctor	Advise customer to return for pharmacist advice, or visit a doctor if pain is not relieved or pain comes back

Assessment questions for the pharmacy assistant

Select one correct answer from each of the following questions.
Answers due 31 May 2012.

Use the Counter Connection module to answer the following questions. Photocopy and/or use the answer sheet provided. Make sure to include your ID number or go online to submit your answers at: **www.psa.org.au/selfcare**

The pass mark for each module is five correct answers. Participants receive one credit for each successfully completed module. On completion of 10 correct modules participants receive an Achievement Certificate.

Submit answers online

To submit your response to these questions online, go to the PSA website: **www.psa.org.au/selfcare**

1. Choose the INCORRECT statement about pain.

a. Pain can be a dull ache or cramp or sharp, stabbing or throbbing pain.
b. Different people can tolerate different degrees of pain.
c. Chronic pain tends to go away as the body heals.
d. Emotional stress and fatigue can make pain feel worse.

2. Choose the INCORRECT statement.

a. Neuropathic pain is often described as tingling, burning, shooting or electric-type pain.
b. Mild to moderate acute pain can be managed with non-prescription pain relievers.
c. Pain that is affecting a person's sleep, mood or concentration requires referral to the pharmacist.
d. Non-prescription pain relievers are generally effective for treatment of neuropathic pain.

3. Choose the INCORRECT statement about NSAIDs.

a. Gastrointestinal side effects are more likely when NSAIDs are used for longer than 7–10 days.
b. Using more than one NSAID at a time provides better pain relief without any risk of side effects.
c. Aspirin is not recommended for use in children under 16 years of age unless directed by a doctor.
d. Ibuprofen is the only oral NSAID recommended for use in children under 5 years of age.

4. When is it necessary to refer to the pharmacist?

a. A mother asks about the dose of paracetamol to give to her three-week old daughter.
b. A woman is having her wisdom teeth out next week and wants to take aspirin for pain relief in the mean time.
c. A 69-year-old man has been taking ibuprofen every morning for a couple of months for the pain and stiffness in his knees.
d. All of the above.

5. Choose the CORRECT statement.

a. Oral and topical NSAIDs can be recommended to be used together without medical advice.
b. People taking low dose of aspirin to prevent blood clots should only use NSAIDs on medical advice.
c. About 50% of people with asthma experience worsening asthma symptoms or an asthma attack after taking NSAIDs.
d. NSAIDs can be recommended for self-treatment of pain for up to one month without medical supervision.

6. Referral to a pharmacist is generally necessary for which of the following customers?

a. Has had pain for more than three months.
b. Is pregnant or breastfeeding.
c. Requests a codeine-containing pain reliever.
d. All of the above.

Congratulations to Glenda

By Megan Del Dot, Branch Support Pharmacist Queensland

Our congratulations go to Glenda Ricker from the Friendlies Societies Dispensary in Gympie. In February, a delighted Glenda received her seven year Counter Connection Certificate.

Glenda has worked in community pharmacy for 10 years. She loves her job and thoroughly enjoys interacting with her customers of all ages. Glenda explained that working together as part of a highly dedicated team at the Friendlies Society Dispensary makes her look forward to coming to work each day.

Glenda finds the Counter Connection modules interesting and a valuable resource that helps to keep her constantly up to date about different conditions people come to the pharmacy seeking advice about. Glenda feels that the modules act to reinforce her learning every day.

Glenda Ricker

Increasing confidence

By Angela Tuddenham, Field Officer Victoria

I had the pleasure of presenting Thea Dimopoulos her Year 7 Certificate of Achievement. Thea was unaware that I was coming in and was thrilled at her certificate presentation.

Thea has worked at the Lalor Central Pharmacy since it opened in 2001. With more than 30 years pharmacy experience, Thea finds her work rewarding and enjoys meeting new people.

She believes Self Care is terrific and finds the material in *inPHARMation* very interesting and enjoys learning something new each month. Her increasing knowledge gives her confidence when dealing with customers. Congratulations Thea!
We look forward to seeing you at your 10 year presentation!

Thea Dimopoulos

10 wonderful years of Self Care

By Lisa Parker, Pharmacy Professional and Support Services South Australia

Retail Manager Rosa Sabatino and Pharmacist Li Yeen Goh

United Discount Chemists Wattle Park in South Australia celebrated its 10th anniversary of being a Self Care member on 4 March.

In recognition of this fantastic achievement, a special afternoon tea was held at the pharmacy where we enjoyed some delicious bakery treats and freshly brewed hot coffee – perfect for a wet and wintery Adelaide day!

Wattle Park is located at the base of the Adelaide Hills, a 10–15 minute drive from the city centre. Despite being in the Adelaide metro area, there is a real country community vibe when you arrive at Wattle Park, and this feeling is further encapsulated in the local shopping centre where the pharmacy is situated.

I was greeted with a warm welcome by Retail Manager, Rosa Sabatino, as soon as I stepped foot inside the pharmacy. Rosa has been working at Wattle Park Pharmacy for more than five years and after spending only a few minutes with her, you can tell she is very passionate about what she does. Rosa has a vibrant personality and exudes enthusiasm and energy, which is highly contagious and has a positive affect on both her fellow staff members and the customers.

During Rosa's time at the pharmacy she has seen the store undergo a number of big changes, including a change of banner group and more recently, the relocation to a new premises at the start of this year. The new store, which is 'bigger, brighter and better', includes a seated counselling area, where the Self Care Fact Cards are on hand and ready for the pharmacists to use in patient counselling.

The pharmacy assistants also make great use of the Fact Cards and frequently hand them out to customers as an extra source of product information. *Head lice, vaginal thrush, anti-fungal* and *smoking cessation* are particularly popular titles, as 'customers are always wanting more information on these difficult conditions'.

The staff also enjoy completing the monthly Counter Connection modules, which provide them with in-depth training across a wide variety of topics. All staff agreed that it was particularly important that their knowledge was kept up to date with continual training. The proof is certainly in the pudding, as I was able to witness their extensive knowledge shine through in their customer service, which is a testament to their dedication and commitment to the Self Care program.

Congratulations again to the team from Wattle Park on achieving 10 fabulous years with Self Care. It has been an absolute pleasure having you with us and we would like to wish you all the best as you settle in to your new pharmacy.

The new counselling area, complete with Self Care Fact Cards

Conferences and calendar dates

Conferences

37th PSA Offshore Refresher Conference

27 April – 3 May, 2012

Istanbul, Turkey

Contact: info@impactevents.com.au

National Medicines Symposium 2012

24–25 May, 2012

Sydney Convention and Exhibition Centre
Sydney, NSW

www.nps.org.au

PSA Clinical and Practice Expo

26–27 May 2012

Hordern Pavillion, Sydney, NSW

Website: www.psa.org.au

ConPharm 12

1–3 June, 2012

Hyatt Regency Coolum, Sunshine Coast, Qld

Registrations open January 2012

www.aacp.moodle.com.au

The Pharmacy Management Conference

29 August – 2 September 2012

Hyatt Regency Coolum, Sunshine Coast Qld

Email: pharmacyconference@acclaimsemm.com.au

National health calendar dates

April	
25	***ANZAC Day***
May	
1	***World Asthma Day*** Asthma Foundation www.asthmafoundation.org.au
2–8	***National Mothering Week*** Australian Breastfeeding Association www.breastfeeding.asn.au
6–12	***Heart Health Week*** Heart Foundation www.heartfoundation.org.au
12–16	***Myeloma Awareness Week*** Leukaemia Foundation www.leukaemia.org.au
25	***Australia's biggest morning tea*** Cancer Council Australia www.biggestmorningtea.com.au
31	***World No Tobacco Day***

What's coming up in *inPHARMation*

The next issue of *inPHARMation* will focus on 'Heart health'. Pharmacists and pharmacy staff can help to raise awareness about the warning signs of heart attack. They can also provide important advice about what people should do if they develop warning signs and why it's important to act quickly. This issue will also discuss the risk factors for coronary heart disease, and the medicines and lifestyle modifications used to manage these risk factors and reduce the risk of heart attack.

Subscribe to *inPHARMation*

Self Care members receive two copies of *inPHARMation* every month as part of their annual subscription. Your staff can have their own copy mailed directly to them at home or work.

Additional *inPHARMation* 12-month subscription $50.00 inclusive of GST.

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PO Box 42 DEAKIN WEST ACT 2600
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Self Care Fact Cards

Please make sure that you keep your Fact Cards up to date. You can re-order any Fact Card title at any time. Simply ring **02 62834777** or fax order to **02 62852869** or go to Self Care at **www.psa.org.au/selfcare**

Fact Cards are included in your membership.

Self Care contacts

National Office Ph: 02 6283 4777

Membership Enquiries Ph: 1800 303 270

Manager, Practice Programs Deborah Cracknell

Program Coordinator Kylie Davis

Professional Programs Pharmacist Lynn Greig

Practice Pharmacist Carolyn Allen

To contact your local Branch phone **1300 369 772**

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TAKING THE PAIN OUT OF COLD & FLU

You are probably used to recommending products that combine phenylephrine and paracetamol to treat the symptoms of cold and flu. Now there is a new combination of ibuprofen and phenylephrine available, what else do you need to know?

Why is ibuprofen an appropriate recommendation for the relief of cold and flu symptoms?

The symptoms of cold and flu vary, but there are some that occur regularly. Nasal congestion is the most common, affecting 90% of people. Body aches and pains are also common, affecting 2 out of 3 cold and flu sufferers.[1]

Why do so many people with cold and flu experience aches and pain? It's because colds and flu are caused by viral infections. Infection, and the immune system's response to infection, can result in the production of chemical mediators called prostaglandins. The release of prostaglandins can lead to aches and pains around the body, as well as inflammation in muscle tissue and in the mucus membranes of the nose.[2,3]

That's where ibuprofen comes in.

It targets the source of the pain by blocking the production of prostaglandins, resulting in significantly reduced muscle and joint pain due to cold and flu.[3]

New Nurofen Cold & Flu PE combines pain-reducing ibuprofen with phenylephrine for the effective relief of cold and flu symptoms.

Next time a customer comes to you with cold or flu symptoms, recommend Nurofen Cold & Flu PE to reduce their body aches and pain and nasal congestion.

Want to know more about cold and flu?

Colds and flu are among the most frequently cited reasons that customers come into pharmacies,[4] so it is important that you know all about this topic and the products that are available, in order to make the best recommendation.

To help you train, we are developing a new Cold & Flu module for PainPod, the online training programme from Nurofen that mixes interactive case studies, animations and quizzes, to make training enjoyable. The existing pharmacy assistant modules 1–6 also count as QCPP Approved Refresher Training!

APPROVED REFRESHER TRAINING
Supporting Excellence in Pharmacy

Register at www.painpod.com.au to start training today!

References: 1. Eccles R *et al. Curr Med Res Opin* 2006; 22(12): 2411–8. 2. Eccles R. *Lancet Infect Dis* 2005; 5(11): 718–25. 3. Winther B and Mygind N. *Am J Rhinol* 2001; 15(4): 239–42. 4. IMS HCP quantitative market research, January 2010.